

Practical Cyber Attack Detection With Continuous Temporal Graph in Dynamic Network System

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Abstract—Deep learning (DL) greatly enhances cyber anomaly detection capabilities through effective statistical network characteristic. However, previous methods have not fully addressed two real-world scenario-driven challenges. 1) Frequent node access and disconnection sourced from free-bounded 5G/B5G cyberspace introduce unfamiliar communication behavior patterns, reducing the detection ability of the pre-trained DL model. 2) Low-frequency or sporadic communication behaviors lack stable patterns, posing a challenge for existing AI-driven models, including DL-based detection methods. To address these issues, we propose a cyber anomaly detection framework based on Continuous Temporal Graph (CTG) neural network from a new interaction-centered perspective. The proposed framework refines the concrete information interaction between network entities into the CTG evolution process, thereby naturally incorporating new node access behaviors into feature extraction on CTG neural network. We furthermore present a message aggregation scheme on CTG with fusion of spatio-temporal neighborhood, the actual time distribution and the historical state, thus transforming communication into a more stable pattern for the learning of low-frequency interactions. Extensive experiments on 4 novel datasets, including ToN-IoT, UNSWNB15, CIC-Dark2020, J.P. Morgan payment, demonstrate that our approach outperforms state-of-the-art methods, particularly in detecting new access and low-frequency behaviors.

Index Terms—Graph neural network, intrusion detection, anomaly detection, semisupervised learning.

I. INTRODUCTION

IT IS estimated that by the end of this year, 5.3 billion people and 29.3 billion devices will be interconnected by prospering networking technologies such as Beyond Fifth-Generation (B5G) and the Industrial Internet of Things (IIoT) [1]. Consequently, people, machines, and things are more closely interacting than ever before. The rapid growth of networked devices and a significant increase in interactive

data are driving the attack surface to inflate rapidly. Therefore, the malicious behaviors of cybercriminals can be easily blended into these complex, massive, and (especially) multi-source interactive data, just like a needle in a haystack. Network threats mixed with frequent interactions, i.e., phishing campaigns, unsolicited cryptomining, ransomware, and information-stealing malware, are intensively becoming a systemic challenge. Therefore, the detection of anomalous network behaviors among massive interactive data plays an essential role in ensuring the sustainability and security of cyberspace.

Extensive research has been conducted to analyze and classify various types of abnormal cyber behaviors, e.g., expert systems, signature-based network behavior analysis approaches, modern machine learning (ML)-based methods and advanced deep neural network models. However, most of these techniques focus on behavioral attributes rather than interactive processes. In reality, these methods based on behavioral feature representation often exhibit limited effectiveness due to some unavoidable factors, such as anonymity technology and privacy protection. For instance, signature-based methods cannot cope with encrypted data [2], while the additional regulatory and privacy requirements within financial services usually disclose limited historical data and few available features [3].

Furthermore, targeting better detection performance, other researchers are attempting to mine temporal and spatial information in cyberspace by employing graph-based learning approaches, e.g., Anomal-E [4], a dynamic line graph [5], a spatial-temporal graph convolutional network [2], a wrongdoing monitor [6], and WTAGRAPH [7]. Each of these graph-based methods contributes to achieving superior data representation capabilities. For the aforementioned successful works, the topological and temporal information encapsulated in network user interactions reflects distinct behavioral characteristics. Understanding these characteristics can aid in uncovering potential security risks. However, several challenges persist in previous methods.

The first limitation is that the extraction of spatial-temporal features is restricted to a fixed topological structure by the existing graph-based methods [4], [5], which hinders the discovery of anomalies from new connections and interactions. Concretely, these methods generally treat the network topology as a consistent adjacency matrix during training and testing, which means that new topological information is difficult to fully learn and express. However, we are facing a

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more complicated and unbounded cyber environment [29]; for instance, attackers may use new virtual hosts to launch DDos attacks, and scammers can utilize new accounts to commit fraud.

The second challenge lies in the inefficiency of most anomaly detection methods when confronted with the non-uniform distribution of network traffic traces across spatial and temporal domains. This inefficiency is particularly pronounced in handling low-frequency communication behavior among network hosts [30], [31]. For instance, statistics from the Ton-IoT, CIC-Darknet, and CIC-IDS2018 datasets show that 5% of all host pairs have contributed more than 50% of the communication behaviors contained in all test sets, while the frequency of communication between the remaining 95% of the host pairs is less than 20 times. Traditional deep learning (DL) methods have difficulty mining stable statistical patterns from these low-frequency sporadic behaviors. Additionally, graph-based detection methods such as those of [2] and [5] assume that the spatial topology behind network communication is uniformly distributed in the time dimension while incorporating spatial-temporal characteristics. Specifically, these methods first extract the spatial information of each graph snapshot and adopt sequence models to model the sequence associations between these pieces of spatial information. In the face of these occasional interaction events, the sequence models adopted by [2] and [5] may not capture effective information for numerous time slices.

To address the above issues, we propose a novel adversarial interaction behavior detection framework based on a continuous temporal graph neural network (CTGNN) to capture both the attribute and complex behavior information of network interactions. Specifically, the proposed framework refines the concrete information interaction processes between each pair of entities as corresponding interactive edges on a continuous temporal graph (CTG) so that the behavior generated by a new node access event can be naturally expressed as a CTG update. Furthermore, low-frequency behaviors are no longer processed in isolation but are incorporated into an entire spatiotemporal neighborhood for message aggregation, thus transforming them into a more stable pattern for model learning. By adopting richer cyberspace behavior features during the interaction process, the characteristics of network interactions can be more effectively learned, analyzed and summarized to efficiently distinguish malicious behaviors. The main contributions of this paper are summarized as follows.

- We propose a cyberattack detection framework leveraging a CTG to address the challenge posed by fixed topologies in detecting newly connected entity behaviors. For the first time, the proposed method refines anomaly detection from a macroscopic topology level to a granular level, focusing on concrete information interactions between pairs of network entities. This approach naturally accommodates new node access behaviors within feature extraction.
- To independently track the long-term historical status of the interaction process, we develop a CTGNN model. Unlike previous methods that primarily rely on sequence relationships to capture historical information [8], our CTGNN model offers a more detailed analysis by incorporating information updates

and topological changes between network entity pairs. This enriched representation capability enhances the effectiveness of the detection model.

- We propose a message aggregation scheme based on spatiotemporal neighborhoods to mitigate the impact of low-frequency interactions. Our method systematically integrates interaction behavior details, including contextual neighborhood representations, temporal distributions, and historical states. This comprehensive approach improves detection performance, particularly for occasional interactions.
- Extensive experiments are conducted on 4 datasets covering different interactive scenarios, including intrusion detection in the IIoT [9] and the intranet [10], J.P. Morgan fraud detection in transactional networks [3] and darknet traffic analysis [11]. The obtained results demonstrate that the proposed method significantly outperforms the state-of-the-art approaches.

This paper is structured as follows. The related works are reviewed in Section II. Section III provides the formal problem definition. Section IV describes the employed methodology. In Section V, extensive experiments and results are presented. Section VI is devoted to the conclusions and further work.

II. RELATED WORK

A. Abnormal Detection in Interactive Network Behavior

The detection and mitigation of abnormal interaction behaviors have long been prominent trends in the field of cybersecurity, including anomaly detection and darknet analysis in network traffic. Different from early rule-based methods such as Snort [12], ML- and DL-based methods have been widely explored to better detect malicious cyber activities with favorable scalability and high accuracy [13]. For instance, Marteau [14] proposed a semisupervised anomaly detection approach based on binary trees with random partitioning. Bitton and Shabtai [15] proposed an anomaly detection technique based on PCA and K-means for detecting malicious TCP packets, which can carry exploits aimed at a remote desktop server.

Compared with above mentioned ML-based methods, flexible deep architectures and end-to-end learning processes enable DL-based methods to capture more sophisticated attack patterns. Yang et al. [16] developed a conditional variational autoencoder with extreme value theory to perform unknown intrusion detection by leveraging flow-level features. Kravchik and Shabtai [17] examined a cyberattack detection method for industrial control systems based on 1D convolutional neural networks (CNNs) and autoencoders. Moreover, Zhao et al. [8] presented an error-resilient recurrent neural network (RNN) for network-induced phenomena, thereby effectively detecting encrypted traffic. The above methods emphasize explicit attribute features, while implicit behavioral and topological features are difficult to deal with.

B. Graph-Based Abnormal Interactive Detection

Recently, several pioneering works have intensively explored NIDSs based on GNNs to exploit the spatial information of actual networks to varying degrees. For instance,

Caville et al. [4] presented a static graph-based intrusion detection method that leverages edge features and a graph topology structure in a self-supervised manner. Moreover, Wang and Zhu [6] integrated the inter- (event-level) and intra- (property-level) associations of behaviors into a unified graph for anomaly detection purposes. Furthermore, temporal information concerning topological structures has attracted researchers' attention. Among them, Cao et al. [2] mapped a data plane-programmable SDN to a graph and deployed a spatial-temporal graph convolutional network to detect DDoS attack flows. Similarly, Duan et al. [5] converted network traffic into a series of line graphs and employed a dynamic line GNN-based NIDS to achieve improved intrusion detection performance. However, the underlying GNNs in the literature [2], [4], [5], [6] are all traditional static GNNs, which are limited to invariant topological structures. In other words, it is difficult for these methods to satisfy actual network environments, such as 5G NFV or 6G space-air-ground integrated networks, with time-varying characteristics, unbounded cyber environments and diverse connection densities.

C. Dynamic Graph Representation Methods

Dynamic graph representation is an emerging DL paradigm for learning evolving relationships among various entities across temporal dimensions. In general, dynamic graphs can be broadly classified into two categories: discrete-time dynamic graphs (DTDGs) and continuous-time dynamic graphs (CTDGs). The evolution process of a DTDG can be described by a sequence of static graph snapshots sampled at regularly spaced times. Actually, the spatiotemporal graph and dynamic line graph representations used in the above anomaly detection methods [2], [5] can be regarded as different types of DTDGs.

Different from a DTDG, a CTDG is modeled as a stream of serial events that occurs sequentially and has a finer temporal granularity. Zuo et al. [18] mentioned that snapshot-based methods (such as [2] and [5]) model temporal correlations as a fixed topological structure, thus ignoring the fine-grained topological variation process. Reference [18] suggested utilizing the neighborhood formation process to describe the evolution of a node and adopting a Hawkes process to generate the corresponding node representation. Ma et al. [19] proposed a novel CTDGNN framework that continuously updates node information by capturing the interaction information and the time intervals of edges. TGAT [20] utilizes a graph-based temporal attention mechanism to capture the temporal neighborhood information of a node, where the positional encoding of the attention mechanism is replaced by a temporal encoding. Based on that, Rossi et al. [21] added a memory module to TGAT to trace the long-term dependencies of different nodes.

However, the above methods impose a certain upper limit on abnormal cyber interaction detection performance. Specifically, typical DGNNs usually focus on node embedding-driven tasks such as temporal link prediction or node classification, whereas abnormal cyber interaction detection tasks aim at the link states and communications between network agents. Moreover, the message aggregation mechanisms adopted by

the previous DGNNs tend to aggregate high-frequency node signals from nodes with more connections, making it challenging to fully represent low-frequency or isolated behaviors.

III. PROBLEM DEFINITION

A. Threat Model

Our abnormal cyber behavior analysis task involves detecting malicious network behaviors by understanding their underlying interactive communication processes. Assuming an unbounded network environment, such as the Internet of Things, 5G or a trading network, would allow possible entities access and induce disconnections. Additionally, considering the diverse connection density characteristics of actual networks, some communication-intensive nodes possess frequent interactions, while other inactive entities only send a small amount of information. Malicious actors can exploit network entities to carry out various malicious activities, including DDoS attacks, brute-force attacks, money laundering, and fraudulent transactions.

We assume that security analysts are capable of recording the interactive behaviors between cyber entities. Taking the intrusion detection scenario as an example, the network traffic interaction can be recorded in the network flow format, which is indexed by the following default five-tuple: `<Source IP, Source Port, Destination IP, Destination Port, Protocol>`. These recordings can represent the protocol-related, size-related and time-related basic features of the interaction between the server host and the network switch. In addition to the attribute features in the traditional network security analysis process, we combine the topology, interaction, and timing features between interactive events to introduce more information, which provides a new CTG-based modeling paradigm for cyber anomaly detection.

B. Continuous Temporal Graph

An interactive CTG is defined as a directed multigraph $G = (V, E, Event)$, where V represents an initial node set, and each $e \in E$ denotes a communication link between pairs of nodes. Each $event \in Event$ represents a bipartite interaction event between nodes that corresponds to a network traffic flow or transaction record. Since multiple interactions directed from host u to host v are permitted in network G , we use different timestamps $event_{u,v}^t$ to indicate the interactions among the links (u, v) . Moreover, when a new node i accesses the network and communicates with node u , a corresponding interaction event $event_{i,u}^t$ is generated. Intuitively, the CTG can represent the interaction processes and temporal associations in a time-varying network. As shown in Fig. 1, each node denotes a host IP, while each interaction represents a specific netflow issued from the source host to the destination host. In line with real network characteristics, multiple bidirectional events occur between nodes, each representing different network interaction records.

C. CTG-Based Interaction Representation and Classification

With the above definition, the task of abnormal cyber interaction detection is formally converted to a classification

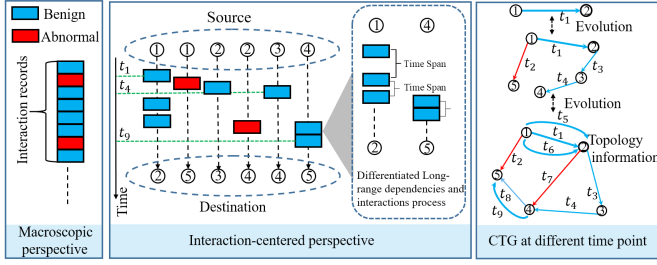


Fig. 1. An example of the traffic temporal graph constructed from UNSW-NB15 network traffic.

problem in the proposed CTG. Following the terminology in [22], we employ an encoder-decoder pair to resolve the interaction event representation learning and classification problem, where the encoder maps all interaction events to corresponding edge embeddings on the CTG. Formally, given a CTG $G = (V, E, Event)$, the goal is to design a temporal graph encoder $f_\omega : x^{n \times e} \rightarrow \mathbb{R}^{n \times d}$ ($d \ll e$) that generates a d -dimensional representation for n input events with e -dimensional interaction features in G . Furthermore, a decoder takes interaction representations as inputs and outputs the prediction $\hat{y}$, where each component of $\hat{y}$ represents the conditional probabilities of each class. Specifically, our tasks in this work include classifying interaction events as benign or malicious and further distinguishing between different attack categories. The learned edge representations can be utilized in different ways, including supervised learning, semisupervised learning, unsupervised clustering, and few-shot learning.

IV. METHODOLOGY

As illustrated in Figure 2, the proposed CTGNN is composed of four components. Specifically, the *temporal graph builder* first constructs a sequence of temporal graphs from interaction records and determines the local event neighborhoods and event features. Next, the *Bochner time encoder* is deployed to map the time span to a d -dimensional vector. Subsequently, the *interactive event update* component has the ability to trace and update long-term dependencies across the corresponding events. The *interaction event aggregation scheme* fuses interaction behavior details including temporal-topological neighborhood information, the actual time distribution and the historical evolution process of each event.

A. Continuous Temporal Graph Construction

First, we transform the original interactive records into a CTG, as exemplified in Figure 1. Taking the interactive traffic in the UNSW-NB15 dataset as an example, all intercepted traffic flow records are treated as interaction events between corresponding source and destination hosts. With this setting, each node of the CTG represents a specific IP address. The edge features of the CTG are different network flow statistics that depend on the specific flow format and the exported flows, such as the protocol, duration, number of incoming flow bytes, and number of bytes per packet. It should be emphasized that

the proposed CTG does not consist of a fixed adjacency matrix. Referring to [19] and [20], the proposed CTG dynamically updates the topology structure and edge feature through the interaction relationships in chronological order. Particularly, in traffic interaction scenarios, the interaction relationship is determined by the following quadruple: $\langle \text{Source IP, Destination IP, Time, Index} \rangle$.

In traditional GNNs, the local neighborhoods of nodes are naturally defined by explicit edge connections. Notably, in the interactive CTG, both network topology information determined by entities (such as IPs and transaction senders) and contextual relationships between interaction events in the time dimension are included. As shown in Figure 1, for event $e_{u,v}^t$ at time t , we consider its 1-hop neighborhoods $\mathcal{N}(e_{u,v}^t) = \left\{ \left\{ e_{u,1}^0, \dots, e_{u,n}^{t-} \right\} \cup \left\{ e_{1,v}^0, \dots, e_{n,v}^{t-} \right\} \right\} \setminus \left\{ e_{u,v}^0, \dots, e_{u,v}^{t-} \right\}$, where $\left\{ e_{u,1}^0, \dots, e_{u,n}^{t-} \right\}$ indicates the interaction events launched from source node u , $\left\{ e_{1,v}^0, \dots, e_{n,v}^{t-} \right\}$ stands for the interaction events received by destination node v , and $\left\{ e_{u,v}^0, \dots, e_{u,v}^{t-} \right\}$ is the set of historical events occurring between u and v . Such neighborhood settings are mainly based on the following considerations. If an interactive record is a malicious network behavior, then the initiator of the interaction may send malicious messages to other entities. Similarly, other behaviors accepted by the interaction destination of this record may also be malicious behaviors. By defining the local neighborhoods of interaction events, we can further design a corresponding message passing mechanism, thus fusing the temporal-topological information between interaction events.

B. Bochner Time Encoder

Sequence models such as long short-term memory (LSTM) and gated recurrent units (GRUs) can mainly understand contextual relations, but it is difficult for them to model the actual time distribution between interactive events. Therefore, a Bochner time encoder [23] is deployed to extract and express the potential periodicity and translation invariance of the actual time distribution among interactive events. Formally, suppose we are given a function mapping $\Phi : T \rightarrow \mathbb{R}^d$, where T and $\mathbb{R}^d$ are the continuous time domain and a d -dimensional vector space, respectively. Assume that for two time points $t_1, t_2 \in T = [0, t_{\max}]$, the corresponding time kernel can be defined as $\mathcal{K}(t_1, t_2) := \langle \Phi(t_1), \Phi(t_2) \rangle$ and $\mathcal{K}(t_1, t_2) = \psi(t_1 - t_2), \forall t_1, t_2 \in T$ for some $\psi : [-t_{\max}, t_{\max}] \rightarrow \mathbb{R}$. We adopt the Bochner time encoder [23] to map the time interval to a d -dimensional vector.

According to Bochner's Theorem, this time kernel $\mathcal{K}$ could be expressed as follows:

$$\begin{aligned} \mathcal{K}(t_1, t_2) &= \psi(t_1, t_2) \\ &= \int_{\mathbb{R}} e^{i\omega(t_1-t_2)} p(\omega) d\omega = \mathbb{E}_\omega [\xi_\omega(t_1) \xi_\omega(t_2)^*] \end{aligned} \quad (1)$$

where $\xi_\omega(t) = e^{i\omega t}$. Since the probability measure $p(\omega)$ is a real-valued function, the real part of equation 1 is denoted as:

$$\begin{aligned} \mathcal{K}(t_1, t_2) &= E_\omega [\cos(\omega(t_1 - t_2))] \\ &= E_\omega [\cos(\omega t_1) \cos(\omega t_2) + \sin(\omega t_1) \sin(\omega t_2)] \end{aligned} \quad (2)$$

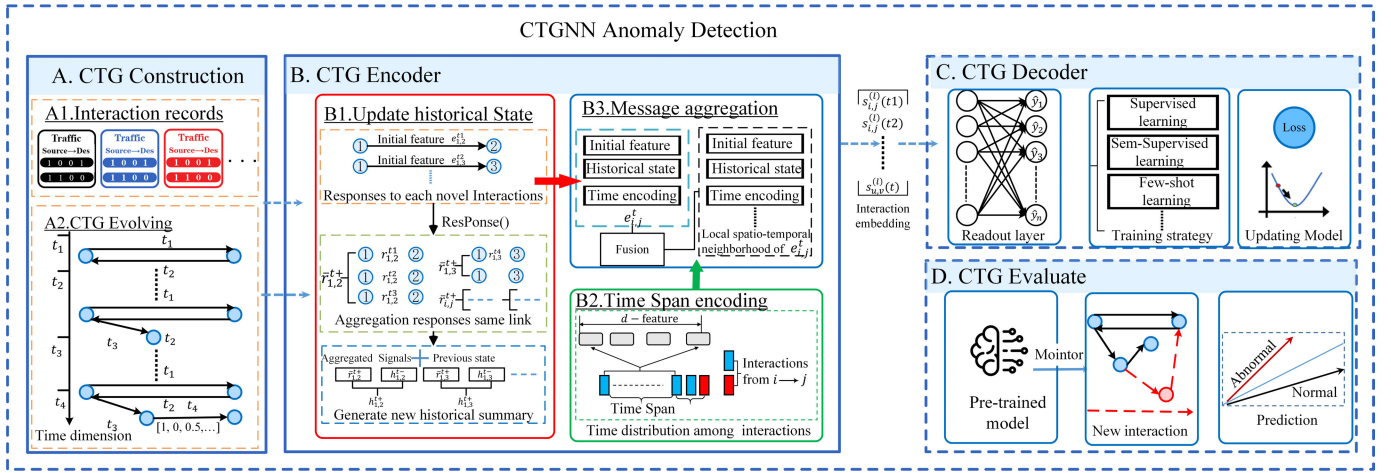


Fig. 2. The framework of the dynamic-graph-based NIDS model.

The above expectation can be approximated by Monte Carlo integral. Assuming $\omega_1, \dots, \omega_d \stackrel{\text{i.i.d}}{\sim} p(\omega)$, the time kernel $\mathcal{K}(t_1, t_2)$ could be estimated as $\frac{1}{d} \sum_{i=1}^d (\cos(\omega_i t_1) \cos(\omega_i t_2) + \sin(\omega_i t_1) \sin(\omega_i t_2))$. Therefore, Bochner's theorem motivates the time feature mapping $t \mapsto \Phi_d^B(t)$ as:

$$\sqrt{\frac{1}{d}} (\cos(\omega_1 t), \sin(\omega_1 t), \dots, \cos(\omega_d t), \sin(\omega_d t)) \quad (3)$$

In our follow-up design, we combine the Bochner time encoder and GRU to model the time distribution of interaction events in a more fine-grained manner.

C. Interactive Event Update Mechanism

To understand differentiated interaction processes between different entities, we model their fine-grained behavioral patterns from an interaction perspective. Specifically, we propose an interactive event update mechanism to independently track and summarize the historical evolution of interaction events between correspondin entities. The interactive event update mechanism maintains a historical state list comprising a set of vectors that summarizes the history of events between different entities. Whenever a subsequent event occurs, its historical state will be updated by the GRU and Bochner time encoder.

Formally, the historical state of an interaction event $e_{u,v}^t$ consists of a vector $h_{u,v}^t$ with d -dimensional features summarizing the historical evolution of $e_{u,v}^t$. It should be emphasized that since the CTG is a directed graph, we allocate separate historical states $h_{v,u}^t$ for $e_{v,u}^t$. The historical state $h_{u,v}^t$ is updated when the subsequent event $e_{u,v}^{t+}$ occurs, where $t+$ denotes a time stamp after t . For each subsequent event $e_{u,v}^{t+}$, a response signal can be computed as follows:

$$r_{u,v}^{t+} = \text{ResPonse}(h_{u,v}^t \parallel \Phi(t^+ - t) \parallel e_{uv}^{t+}) \quad (4)$$

where $\parallel$ indicates the concatenation operation, $\Phi(\cdot)$ represents the Bochner time encoder that captures the time distribution, and e_{uv}^{t+} denotes the feature vector of subsequent events. Additionally, ResPonse signifies learnable functions that merges the current state, the compressed historical state, and the actual

time span feature. Multiple choices could be considered for the learnable function such as MLP, CNN, Transformers, our model selects MLP as the learnable function.

To effectively reduce the computational overhead, the historical state is updated with a batch cycle. We deploy an aggregation mechanism to handle the corresponding responses when multiple events occur on a specific link. Let $\mathcal{J}(e_{u,v}^{t_1}, \dots, e_{u,v}^{t_n})$ denote the events on links (u, v) in a batch; the aggregation operation can be written as

$$\bar{r}_{u,v}^{t+} = \text{agg}(r_{u,v}^{t_1}, \dots, r_{u,v}^{t_n}) \quad (5)$$

Note that we select the average of all responses as the aggregation operation. It should be emphasized that the above aggregation operation is also performed on other interaction event sets of different links, such as (u, j) and (v, j) . Thereafter, the historical state $h_{u,v}^t$ is updated by applying

$$h_{u,v}^{t+} = \text{Update}(h_{u,v}^t, \bar{r}_{u,v}^{t+}) \quad (6)$$

Here, we deploy a GRU as the update function. The new embedding of the historical state $h_{u,v}^{t+}$ is refined by incorporating historical information concerning interactive events, the time distribution and context relationships.

D. Interaction Event Contexts Aggregation

Typically, GNNs generate node embeddings in a simple homogenous graph by aggregating neighborhood information and graph topology features for tasks such as node classification, node clustering, and link prediction. For the task of network attack detection, we employ an event aggregation mechanism to fuse the neighborhood information of events and generate a embedding representation for each event at any time t . Specifically, the temporal representation $\mathbf{s}_{u,v}^{(\ell)}(t)$ of event $e_{u,v}^t$ at layer ℓ can be calculated as

$$\begin{aligned} \tilde{\mathbf{s}}_{u,v}^{(\ell)}(t) &= \text{AGG}^{(\ell)}(\mathbf{s}_{i,j}^{(\ell-1)}(t') \parallel \Phi(t - t')) \\ \mathbf{s}_{u,v}^{(\ell)}(t) &= \text{MLP}^{(\ell)}(\mathbf{s}_{u,v}^{(\ell-1)}(t) \parallel \tilde{\mathbf{s}}_{u,v}^{(\ell)}(t)), \end{aligned} \quad (7)$$

where $\mathbf{s}_{i,j}^{(\ell-1)}(t')$ represents the temporal representations of neighbor evnets $(e_{i,j}^{t'}) \in \mathcal{N}(e_{u,v}^t)$ at layer $\ell - 1$, $\Phi(t - t')$

denotes the time span feature between neighbor events and origin events. There are multiple implementations of the AGG operation, including parameterized functions like MLP, Multi-Head Attention, Informer, et al. In our practice, we adopt a multi-head attention mechanism to model the impact of different neighbors on event embeddings. The AGG operation based on the multi-head attention can be computed as follows:

$$\begin{aligned}\tilde{s}_{u,v}^{(\ell)}(t) &= \text{MultiHeadAttention}^{(l)}(q^{(l)}(t), K^{(l)}(t), V^{(l)}(t)) \\ q^{(l)}(t) &= s_{u,v}^{(\ell-1)}(t) \parallel \Phi(0) \\ K^{(l)}(t) &= V^{(l)}(t) \\ &= [s_{u'_1, v'_1}^{(\ell-1)}(t'_1) \parallel \Phi(t - t'_1), \dots, s_{u'_N, v'_N}^{(\ell-1)}(t'_N) \parallel \Phi(t - t'_N)]\end{aligned}\quad (8)$$

where the initial representation of each event $\tilde{s}_{u,v}^{(\ell)}(t) = (e_{u,v}^t \parallel h_{u,v}^t)$ exploits both the compressed historical state $h_{u,v}^t$ and the current event feature $e_{u,v}^t$. The query $q^{(l)}(t)$ denotes a reference event, and the keys $K^{(l)}(t)$ and values $V^{(l)}(t)$ denote the information of its neighbors. In this case, we adopt scaled dot-product attention to calculate the attention scores of different neighbor events and their temporal distribution, although in principle, any other type of attention mechanism can be utilized. Following the typical GNN message passing mechanism, an MLP aggregates the neighborhood information and the reference event representation.

E. Traffic Events Classifier

To adapt topology-varying network environments, the proposed CTGNN is designed to support any topology during the test phase, including node connections and disconnections. We initialize new memory structures for nodes and interaction events that are not observed during training. The embeddings of the new interaction events and their memory are calculated by the trained components. During the training phase, all computational components (i.e., all learnable functions) involved in the embedding calculation are expected to contribute to the gradient update. Thus, the model first updates the events' memories by compressing and aggregating the historical signal at the beginning of each subbatch and then combines the updated memories and neighborhood features to infer the embeddings of the training events. As a result, the gradient update process depends on all computational components ever calculated. We summarize the training process of the proposed model in Algorithm 1.

In addition, we adopt two different training strategies to test the performance capabilities of our CTGNN. The difference between them is that the fully supervised process uses all event labels from the training dataset, while the semisupervised process only utilizes parts of the training dataset labels. We combine the representations of temporal events into a classification subnetwork that includes a softmax function and minimizes the cross-entropy loss for event classification tasks. Furthermore, in the evaluation phase, we calculate the class probabilities of the test samples and compare them with their actual labels to assess the performance of the model.

Algorithm 1 The Overall Process of the Proposed Method

Input: Interaction events with time steps: $e_{1,2}^0, \dots, e_{u,v}^t$; Corresponding feature matrices $X \in \mathbb{R}^{n \times e}$

Set hyperparameters: Number of events in each batch, number of events in the neighborhood of a specific interaction, labeled data percentage, dropout probability

Output: Prediction $\hat{y}$

- 1: Divide interaction events into batches and establish a CTG, initialize historical state to 0
- 2: **while** not converged **do**
- 3: **for** each $batch = 1 : Max_{batch}$ **do**
- 4: Responses for the new events in the current batch $r_{u,v}^{t+} \leftarrow \text{ResPonse}(h_{u,v}^t \parallel \Phi(t^+ - t) \parallel e_{u,v}^{t+})$
- 5: Aggregate the response signals $\text{agg}(r_{u,v}^{t_1}, \dots, r_{u,v}^{t_N})$
- 6: Obtain the updated historical state by aggregated signals and previous state $h_{u,v}^{t+} \leftarrow \text{Update}(h_{u,v}^t, \tilde{r}_{u,v}^{t+})$
- 7: Fuse the features from the local spatiotemporal neighborhood: $\tilde{s}_{u,v}^{(\ell)}(t) \leftarrow \text{AGG}^{(\ell)}(s_{i,j}^{(\ell-1)}(t') \parallel \Phi(t - t'))$
- 8: Compute final embeddings combining neighborhood information $s_{u,v}^{(\ell)}(t) \leftarrow \text{MLP}^{(\ell)}(s_{u,v}^{(\ell-1)}(t) \parallel \tilde{s}_{u,v}^{(\ell)}(t))$
- 9: Compute the classification loss $\mathcal{L}$ on the labeled data
- 10: Gradient descent and model parameter update
- 11: **end for**
- 12: **end while**
- 13: **return** Prediction $\hat{y}$

V. IMPLEMENTATION AND EVALUATION

A. Experimental Implementation and Setup

1) *Datasets and Data Preparation:* We choose 4 newly published public datasets with spatial information to evaluate the proposed methods: ToN-IoT 2021 [9], UNSW-NB 15 [10], Darknet 2020 [11] and J.P. Morgan payments dataset [3]. These 4 datasets cover different application scenarios and follow different data distributions so that the performance of the model can be effectively tested. Specifically, the ToN-IoT and the UNSW-NB 15 datasets were both created by the Cyber Range Lab of UNSW Canberra. Among them, the ToN-IoT dataset was created by a realistic Industry 4.0 network environment incorporating a combination of normal traffic and various types of attack traffic. The UNSW-NB15 dataset is a hybrid dataset consisting of authentic normal and attack records. Its network packet volume is approximately 100 gigabytes, generating 2,540,044 observations with 47 features. The CIC-Darknet2020 dataset is published by the Canadian Institute for Cybersecurity and consists of 134,348 benign samples and 24,311 darknet samples with 78 features. The J.P. Morgan payment dataset contains a large variety of transaction types representing normal activities as well as abnormal/fraudulent activities. The features of the payment data include the number of clients, time duration, etc.

The data preparation phase consists of the following 2 steps. We first convert the network flows into a temporal graph as described above, with each network flow corresponding to an

TABLE I
CHARACTERISTICS INFORMATION OF ALL THE 4 DATASETS

Datasets	Property			Training Graph Property		Testing Graph Property		
	Record Number	Input Feature Dimension	Scenarios	Abnormal Percentage	Nodes number	Abnormal Percentage	Nodes number	New Access Nodes
Ton-IoT 2021	461,043	38	IIoT Traffic	34.85%	14,216	35.26%	5,419	2,529
USNW-NB 15	1,270,023	47	Network Traffic	10.92%	47	19.45%	43	0
CICDarknet2020	141,530	78	Encrypted Traffic	17.19%	10,491	17.15%	6,327	612
Payment dataset	221,562	6	Payment network	9.76%	93,682	9.60%	23,581	23,448

event. Second, preserving the contextual information process, we divide the interactive events into 3 groups in order of their appearance. The first 80% of the interactive events are treated as the training set, and the middle 5% and the last 15% are designated as the validation and test sets, respectively. In combination with the subsequent data preparation process, we further present the details of these datasets in Table I. More detailed information and the corresponding resource list can be found in the website repository¹²³⁴. The operating environment for the experiments includes a 4.7-GHz i7-10710U CPU, 16 GB of RAM, and the 64-bit Windows 10 operating system. Regarding the utilized software versions, we employ Python 3.8, PyTorch 1.9.0, CUDA 11.1, and PyTorch Geometric 2.0.2.

2) *Evaluation Metrics*: Five classic metrics are adopted to evaluate the performance of our NIDS model, namely, the accuracy (ACC), recall (DR), false alarm rate, precision and F1 score [24] metrics. In the multiclassification experiment conducted to identify different attack types, we use the macro-DR and macro-F1 scores as evaluation indicators. They are calculated as shown below:

$$\text{Macro F1} = \frac{1}{n} \sum_{x=1}^n F1_x \quad (9)$$

$$\text{Macro DR} = \frac{1}{n} \sum_{x=1}^n DR_x \quad (10)$$

where Macro F1 and Macro DR are the unweighted means of the corresponding F1 and DR scores of different classes, respectively. It's need to emphasize that macro metrics give equal importance to the majority and minority classes, allowing us to still obtain objective results on imbalanced datasets.

3) *Baselines*: We compare the performance of the proposed method against the following detection methods:

CNN-BiLSTM [25] is a classical DL model that joins a CNN and bidirectional LSTM to analyze network traffic.

Stacked Random Forest [26] is an ensemble ML model that combines the predictions derived from multiple random forest algorithms for the intrusion detection task. We train the classifier based on the *sklearn.ensemble* and *mlxtend.classifier* modules following the setup developed in the literature [26].

CVAE-EVT [16] synthesized conditional variational auto encoder and extreme value theory to boost known/unknown attack performance in a two-stage training. Since the author conducted unknown attack detection in the second stage, we only compare with the first stage of the known attack detection.

TMG-GAN [32] is a generative adversarial network based attack detection method to deal with imbalanced traffic distribution. TMG-GAN first performs data augmentation for tail attack categories. Subsequently, a three-layer dense DNN classifier is used to identify different attack traffic.

MTH-IDS [33] refers to a stacking ensemble model that enhances multi-attack detection by integrating four tree-based algorithms including Decision Tree, Random Forest, Extra Trees, and XGBoost. In particular, MTH-IDS trains these classifiers for initial attack assessment and then uses an additional XGBoost classifier for final decision making.

Anomal-E [4] leverages edge-level and topological information to generate an efficient edge embedding representation for anomaly detection.

B. Experimental Evaluation

In this section, we evaluate the proposed CTGNN under different learning settings. We first present its overall performance, including binary and multiclass classification performance, and conduct a detailed comparison between the CTGNN and the state-of-the-art baseline methods. Second, we systematically evaluate the proposed model and baseline methods in the face of new access node behaviors as well as interactions with low frequency. Third, we perform additional ablation experiments to evaluate the effectiveness and contributions of the key computational components used in our method. Subsequently, we evaluate the performance of our work under a semisupervised setting with fewer label participated.

1) *Intrusion Detection Scenarios*: In the intrusion detection scenario, our main goals include (1) detecting anomalous network traffic in a network and (2) distinguishing different malicious attacks in background traffic for network administrators to respond to. Table II summarizes the binary classification performance achieved by the tested methods on the UNSW-NB15 and ToN-IoT datasets. Overall, the proposed method exhibits remarkable performance in the binary intrusion detection scenario, with higher accuracy, a higher anomalous

¹<https://research.unsw.edu.au/projects/toniot-datasets>

²<https://research.unsw.edu.au/projects/unsw-nb15-dataset>

³<https://www.unb.ca/cic/datasets/darknet2020.html>

⁴<https://www.jpmmorgan.com/technology/artificial-intelligence/initiatives/synthetic-data/payments-data-for-fraud-detection>

TABLE II
BINARY CLASSIFICATION PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART ALGORITHMS

Method	Dataset	Data Format	Accuracy	Recall	Precision	F1-Score	FAR
MLP	UNSW-NB 15	Vector	98.65%	97.16%	95.97%	96.55%	2.84%
CNN-BiLSTM		Vector&Sequence	98.62%	97.45%	95.54%	96.48%	2.54%
Stacked Random Forest		Vector	98.60%	96.26%	96.55%	96.40%	3.74%
CVAE-EVT		Vector	98.54%	97.58%	95.06%	96.30%	2.41%
TH-IDS		Vector	98.79%	97.39%	96.45%	96.92%	2.60%
TMG-GAN		Vector	98.70%	96.70%	96.64%	96.67%	3.30%
Anomal-E		Static Graph	99.12%	96.37%	97.27%	96.55%	3.62%
Proposed Method		Temporal Graph	98.83%	98.00%	96.07%	97.03%	1.99%
MLP	ToN-IoT 2021	Vector	94.25%	94.36%	89.86%	92.06%	5.63%
CNN-BiLSTM		Vector	87.37%	84.85%	80.42%	82.58%	15.14%
Stacked Random Forest		Vector	95.37%	95.65%	91.61%	93.59%	4.35%
CVAE-EVT		Vector	84.16%	73.07%	80.25%	76.49%	26.93%
MTH-IDS		Vector	99.25%	99.15%	98.72%	98.93%	0.846%
TMG-GAN		Vector	96.24%	98.90%	91.19%	94.89%	1.10%
Anomal-E		Static Graph	99.89%	99.94%	99.78%	99.85%	0.059%
Proposed Method		Temporal Graph	99.98%	99.98%	99.94%	99.96%	0.015%

behavior detection rate, and a lower false-positive rate. For example, the detection accuracy, precision, recall, F1 score and FAR of the proposed method on the ToN-IoT dataset are 99.98%, 99.98%, 99.94%, 99.96% and 0.015%, respectively.

Several interesting implications can be derived from these results. Although binary classification is a fairly simple task, the performance of traditional DL methods such as CNN-LSTM and CVAE-EVT on the TON-IoT dataset is not as powerful as expected. We speculate that the relatively worse performance achieved by the traditional DL methods on the TON-IoT dataset may be attributable to their copious amounts of Boolean or string features, such as `dns_query`, `ssl_version`, etc. Furthermore, our proposed CTGNN and the Anomal-E model outperform all of the remaining baseline methods. This implies that by benefiting from implicit topological features and neighborhood information, GNN methods can achieve satisfactory performance in binary intrusion detection scenarios. Moreover, by comparing the message aggregation results of the CTGNN to those of the traditional static GNN, we find that the former achieves better detection performance due to its finer-grained understanding of interaction behavior details. We further verify this point of view in the follow-up multicategory intrusion detection scenario.

In the multicategory malicious attack recognition scenario, more challenges are encountered, including a copious amounts of attack categories and an extremely unbalanced data distribution. We mainly focus on the following indicators in the multiclassification setting: the detection rate, the F1 value of each category and the corresponding weighted macro value for the whole model. Tables III and IV summarize the detection performance achieved by the proposed method and the baseline methods on 16 categories of malicious behaviors in the TON-IoT and UNSW-NB15 datasets, respectively. Overall, the proposed method outperforms all baseline models on most attack categories.

On the ToN-IoT dataset, the proposed method achieves detection rates and F1 scores exceeding 95% for 8 types

of traffic behavior (i.e., Benign, Backdoor, DDoS, DoS, mitm, ransomware, scanning, and Xss attacks). Fig. 3 demonstrates that some of the password attacks in the TON-IoT dataset are misclassified as DoS and injection attacks; this may be related to the similar context interaction characteristics of these attack behaviors. We review the source IPs involved in the password attacks (192.168.1.31, 192.168.1.30), and these source IPs also perform DoS attacks and injection attacks on parts of the same communication links. These findings demonstrate from another perspective that our method can effectively capture the context interaction characteristics of a network. On the UNSW-NB15 dataset, the proposed method and baseline methods exhibit worse performance attributes due to the extremely unbalanced data distribution. In this case, the macro-DR and macro-F1 scores of our method are 60.34% and 51.67%, respectively, which are also better than those of the other baselines. The proposed methods still maintain good detection rates for each category.

2) *Encrypted Traffic Scenario*: In the encrypted traffic classification scenario, we mainly conduct binary classification experiments. Figure 4 presents the performance of the proposed model and the remaining baseline methods. Although the darknet dataset has 78-dimensional attribute features, some baseline methods do not perform as expected, especially the static graph-based methods mentioned in the literature [4]. Static graph-based models may not be a panacea for any scenario. We believe that the poor performance of the Anomal-E methods may be due to the way they model static graphs. Anomal-E maps all interaction processes (IP: ports) to a complete whole graph, but in encrypted traffic scenarios, anonymous technologies such as port forwarding, port randomization and tunneling protocols make it difficult for these methods to capture the spatial characteristics of interactive behaviors. Repeated interaction patterns originating from the same machine, such as video services and chat services, may produce unpredictable results for static graph modeling methods. In contrast, the proposed model can predict the

TABLE III
WEIGHTED AND MACRO METRIC RESULTS OBTAINED ON THE TON-IOT DATASET

Algorithm	MLP		CNN-BiLSTM		SRF		CVAE-EVT		Anomal-E		TMG-GAN		Proposed method	
	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1
Benign	94.51%	94.77%	93.04%	93.86%	98.23%	97.85%	89.48%	87.84%	99.72%	99.53%	93.46%	96.33%	99.96%	99.97%
Backdoor	99.95%	64.81%	99.90%	60.89%	99.16%	99.57%	0.79%	1.57%	98.32%	98.98%	99.95%	64.00%	99.97%	99.95%
DDoS	70.69%	81.83%	92.55%	89.62%	91.90%	91.27%	25.45%	40.50%	79.42%	83.14%	96.47%	96.18%	97.84%	98.30%
DoS	21.11%	33.71%	15.01%	24.29%	95.06%	96.62%	86.00%	42.75%	95.54%	93.73%	20.25%	33.22%	98.96%	91.41%
Injection	90.95%	92.51%	49.21%	62.14%	78.60%	84.60%	86.65%	44.51%	77.77%	76.83%	93.46%	90.59%	92.80%	90.63%
Mitm	25.85%	37.59%	47.80%	54.44%	32.20%	40.12%	2.31%	4.27%	0.63%	1.25%	70.73%	70.05%	98.05%	88.94%
Password	57.43%	69.37%	30.54%	38.09%	92.06%	95.06%	5.86%	11.08%	66.66%	67.64%	64.80%	66.91%	78.42%	85.96%
Ransomware	73.36%	75.59%	84.83%	81.38%	91.38%	84.77%	1.56%	3.07%	93.45%	89.25%	85.98%	85.06%	100.00%	99.91%
Scanning	81.08%	76.55%	78.53%	72.41%	95.60%	96.55%	2.58%	4.85%	95.93%	96.84%	83.73%	76.06%	95.38%	97.22%
xss	89.55%	74.59%	80.33%	70.81%	86.94%	87.15%	64.28%	60.78%	89.38%	89.98%	91.44%	92.57%	78.18 %	99.91%
Macro avg	70.45%	70.13%	67.17%	64.79%	86.11%	87.36%	36.50%	30.12%	79.68%	79.72%	80.03%	75.66%	96.13%	95.22%
Weighted Average	86.78%	86.36%	83.62%	82.82%	95.67%	95.64%	70.20%	66.32%	95.10%	94.97%	88.51%	88.40%	98.37%	98.36%

TABLE IV
WEIGHTED AND MACRO METRIC RESULTS OBTAINED ON THE UNSW-NB15 DATASET

Algorithm	MLP		CNN-BiLSTM		SRF		CVAE-EVT		Anomal-E		TMG-GAN		Proposed method	
	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1
Benign	97.71%	98.84%	99.63%	93.86%	97.79%	98.90%	99.47%	99.46%	98.58%	98.96%	97.47%	98.71%	97.67%	98.82%
Analysis	0.0%	0.0%	0.0%	0.0%	0.15%	0.29%	28.17%	9.21%	9.59%	8.77%	0.29%	0.37%	19.91%	13.01%
Backdoor	1.28%	2.49%	0.0%	0.0%	9.67%	17.53%	29.02%	9.61%	9.69%	16.33%	0.14%	0.28%	4.42%	5.70%
DoS	1.47%	2.88%	0.08%	0.15%	11.51%	19.09%	1.26%	2.44%	15.83%	19.93%	66.10%	46.25%	79.83%	47.73%
Exploits	88.03%	69.28 %	87.54%	69.08%	90.91%	72.07%	60.35%	57.77%	82.96%	65.84%	58.76%	58.74 %	50.66%	61.32%
Fuzzers	42.25%	49.60%	33.92%	44.56%	49.80%	53.55%	8.89%	15.64%	46.58%	45.62%	63.46%	42.72%	84.15%	54.27%
Generic	98.40%	98.84%	98.24%	98.75%	98.68%	99.22%	97.97%	97.77%	98.49%	98.87%	97.58%	98.56%	97.45%	98.71%
Reconnaissance	76.43%	75.38%	75.07%	74.81%	76.99%	84.27%	73.27%	50.97%	72.91%	80.32%	57.85%	45.68%	75.96%	72.09%
Shellcode	47.95%	40.19%	46.59%	42.09%	66.14%	62.38%	16.23%	19.25%	37.27%	36.48%	0.0%	0.0%	60.68%	38.09%
Worms	13.46%	22.22%	0.0%	0.0%	28.57%	41.38%	0.0%	0.0%	26.92%	30.77%	3.85%	5.56%	32.69%	26.98%
Macro avg	46.87%	46.01%	44.11%	42.86%	53.02%	54.85%	41.46%	36.21%	49.88%	50.19%	44.55%	39.69%	60.34%	51.67%
Weighted Average	96.04%	96.18%	96.73%	96.13%	96.36%	96.49%	96.82%	96.65%	96.14%	96.16%	95.06%	95.63%	96.30%	96.23%

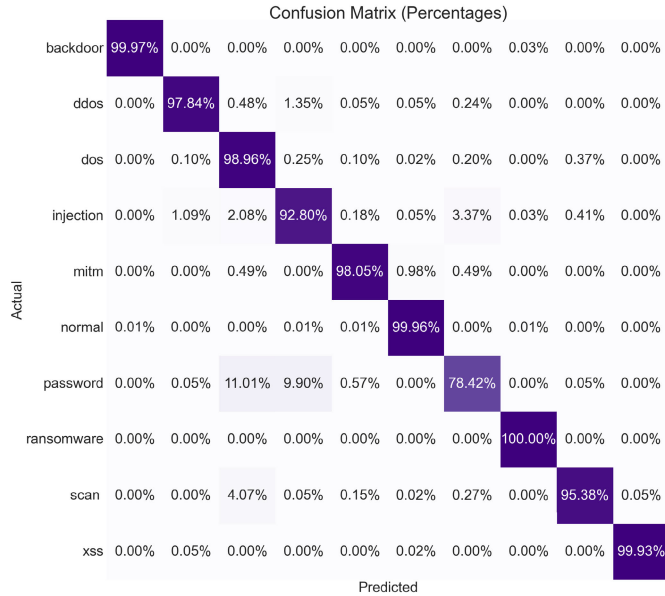


Fig. 3. Confusion matrix of the multiclass classification results obtained on the Ton-IoT 2021 dataset.

141,481 requests of 28,296 testing interaction records in the CIC-Darknet dataset with an accuracy of 99.72%, a detection ratio of 98.99%, a precision level of 98.00%, an F1 score of 97.18%, and an FAR of 1.01%. It becomes evident that our

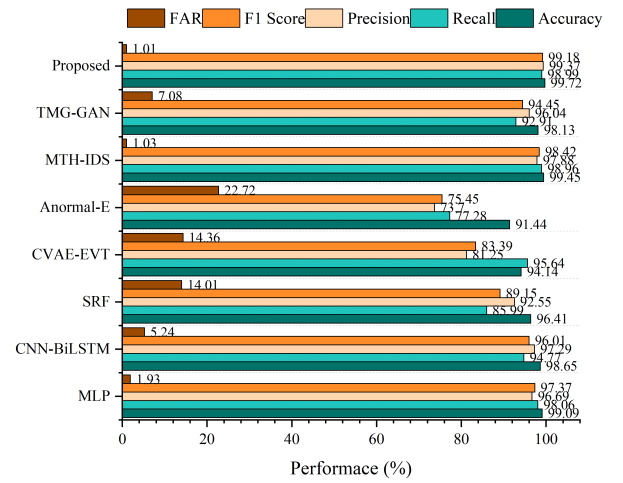


Fig. 4. The performance achieved by the proposed and baseline methods on the CIC-Darknet 2020 dataset.

temporal graph-based models outperform static graph-based models, indicating the power of a finer-grained understanding of interaction behavior details.

3) *Fraud Payment Detection Scenario*: In fraud detection datasets, we are faced with extremely uneven data distributions and quite a few available data features. Fig. 5 shows the fraud detection results of the proposed model and the baseline methods. Notably, the CTGNN consistently outperforms all

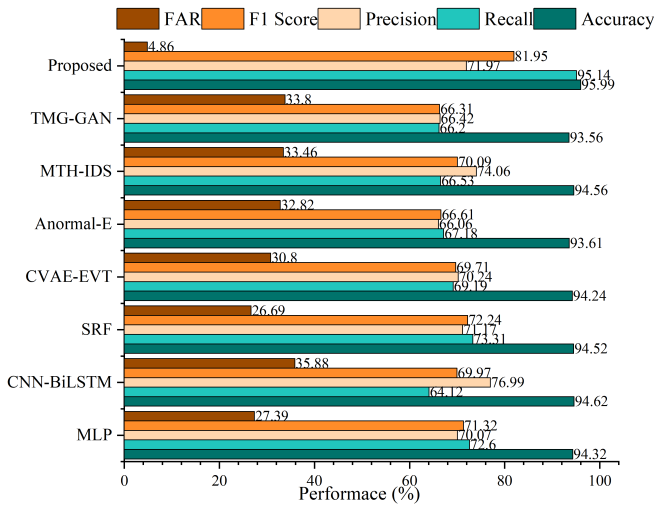


Fig. 5. The performance achieved by the proposed and baseline methods on the Fraud Payment dataset.

baselines with an accuracy of 96.09%, a detection ratio of 81.80%, a precision level of 78.33%, an F1 score of 80.03%, and an FAR of 18.19%. There are two possible reasons for the performance discrepancy exhibited by the baseline methods. First, only 8 features are available, including `USD_amount`, `Sender_Id`, `Sender_Account`, `Transaction_Type`, and `Sender_Country`, in the entire transaction network, so most of the learning methods based on feature representations have worse performance. Second, we discover one limitation of the static graph-based models such as [4]: they may not be the most suitable solutions for these particular highly imbalanced datasets. For example, only 69.71% of the fraud behavior in the J.P. Morgan synthetic transaction network is detected by Anomal-E, while 34.17% of the macro-average DR is reported on the UNSW-NB15 dataset (Table IV). The traditional message passing mechanism reveals the inclination toward learning majority classes. Due to our fine-grained link history modeling method, spatiotemporal interactions, and temporal patterns, our method successfully overcomes the detrimental effects mentioned above in most of our considered datasets.

4) *Analysis of New Access Nodes*: To further verify the impact of behaviors generated by new access nodes on anomaly detection, we conduct a set of manual verifications on the ToN-IoT and CIC-Darknet2020 datasets. As shown in Table I, in the testing phase, 2,529 new access nodes in the IIoT network generate 2,603 interaction records, including normal traffic as well as four types of attack behaviors, i.e., `DoS`, `mitm`, `password`, and `scanning` attacks. In the CIC-Darknet2020 dataset, 612 new interactive IPs generate 820 flow records that occupy 2.89% of the total testing samples. Table V shows the comparison between the results obtained by the proposed CTGNN and the suboptimal baseline methods on these flow records generated by new access nodes. On the ToN-IoT dataset, the performance of the three methods on these newly generated records is worse than that attained on all test samples. To our surprise, SRF does not detect the abnormal behavior generated by the new access nodes.

Anomal-E detects 2 of the 4 types of attacks, while our proposed method detects 3 categories and achieves higher DR and F1 values. On CIC-Darknet2020, facing the encrypted traffic generated by the new access IP, SRF's classification DR and F1 score drop by 12.78% and 11.49%, respectively; Anomal-e's DR and F1 score drop by 16.24% and 9.92%, respectively. In contrast, the proposed CTGNN method still maintains satisfactory performance when confronted with these behaviors.

There are 2 potential reasons for these performance differences. First, we find that 2,544 of the 2,603 newly generated flow records only appeared once, which means that pretrained DL and ML methods are completely unfamiliar to these 2,544 records. The network behavior generated by the new access nodes contains novel feature patterns that are difficult for traditional DL and ML to reason. Although Anomal-E maps all network behaviors to a large graph, these behaviors are isolated nodes in the overall graph structure. Anomal-E does not aggregate enough information to distinguish between different anomalous behaviors. Second, through the aggregation of spatial-temporal neighborhood information, the CTGNN makes these discrete and isolated behavior patterns continuous and compact in the local neighborhood. In other words, the CTGNN incorporates these novel behavior patterns into the local node interaction processes that have been more effectively learned and summarized during the training process, thus better reasoning about these unseen patterns.

5) *Analysis of Behaviors With Low Frequencies*: Noting that most of the behaviors generated by new access nodes occur only once, we further systematically analyze the impact of the frequencies of behaviors on the resulting model performance. Since Anomal-E maps all behaviors to a large graph (all edges appear only once), we compare the CTGNN with SRF on the ToN-IoT dataset. We first count the communication frequency between each pair of IPs in the test set. Specifically, there are 5,680 IP pairs in the test set that produce 92,208 flow records, among which 5,463 pairs of IPs have communication frequencies below 20, generating 10,497 flow records. Furthermore, the frequency of communication between 5,630 IP pairs is less than 300, corresponding to 22,925 flow records. Fig. 6 presents the receiver operating characteristic (ROC) curves produced by the SRF and CTGNN methods, where f denotes different communication frequencies.

As seen from the above figures, although SRF performs well on the overall test samples in Fig. 6a, its performance is not stable when faced with low-frequency traffic records (shown in Figs. 6b and 6c). In fact, Figs. 6a and 6d reveal that the excellent performance of SRF may be mainly concentrated in the behaviors with the majority frequency, the frequencies of these behaviors are greater than 300, but only accounts for 0.88% of the total IP pairs. In contrast, benefiting from utilizing the historical state, the local spatiotemporal neighbor information and the actual time distribution, the proposed CTGNN alleviates the negative impact of the low-frequency interaction distributions. As shown in Figs. 6a and 6d, the proposed approach maintains encouraging performance for most IP pairs, which is essential for real-world scenarios.

TABLE V
PERFORMANCE COMPARISON WHILE CONSIDERING THE BEHAVIORS GENERATED BY NEW ACCESS NODES

Dataset	Categories	SRF-Total		SRF-New		Anomal-E-Total		Anomal-E-New		CTGNN-Total		CTGNN-New	
		DR	F1	DR	F1	DR	F1	DR	F1	DR	F1	DR	F1
Ton-IoT	Benign	98.23%	97.85%	93.86%	99.02%	99.72%	99.53%	99.92%	99.84%	99.96%	99.97%	99.84%	99.88%
	Dos	95.06%	96.62%	0.0%	0.0%	95.54%	93.73%	55.56%	62.50%	98.96%	91.41%	88.89%	76.19%
	Mitm	32.20%	40.12%	0.0%	0.0%	0.63%	1.25%	0.0%	0.0%	98.05%	88.94%	94.74%	80.0%
	Password	92.06%	95.06%	0.0%	0.0%	66.66%	67.64%	0.0%	0.0%	78.42%	85.96%	0.0%	0.0%
	Scanning	95.60%	96.55%	0.0%	0.0%	95.93%	96.84%	14.29%	22.23%	95.38%	97.22%	28.57%	44.44%
CIC-Darknet	Benign	98.57%	97.85%	94.76%	92.94%	95.02%	94.99%	95.64%	93.74%	99.87%	99.83%	100.0%	100.0%
	Encrypted	85.99%	89.15%	73.21%	77.66%	75.64%	75.77%	59.40%	65.85%	98.99%	99.18%	100.0%	100.0%

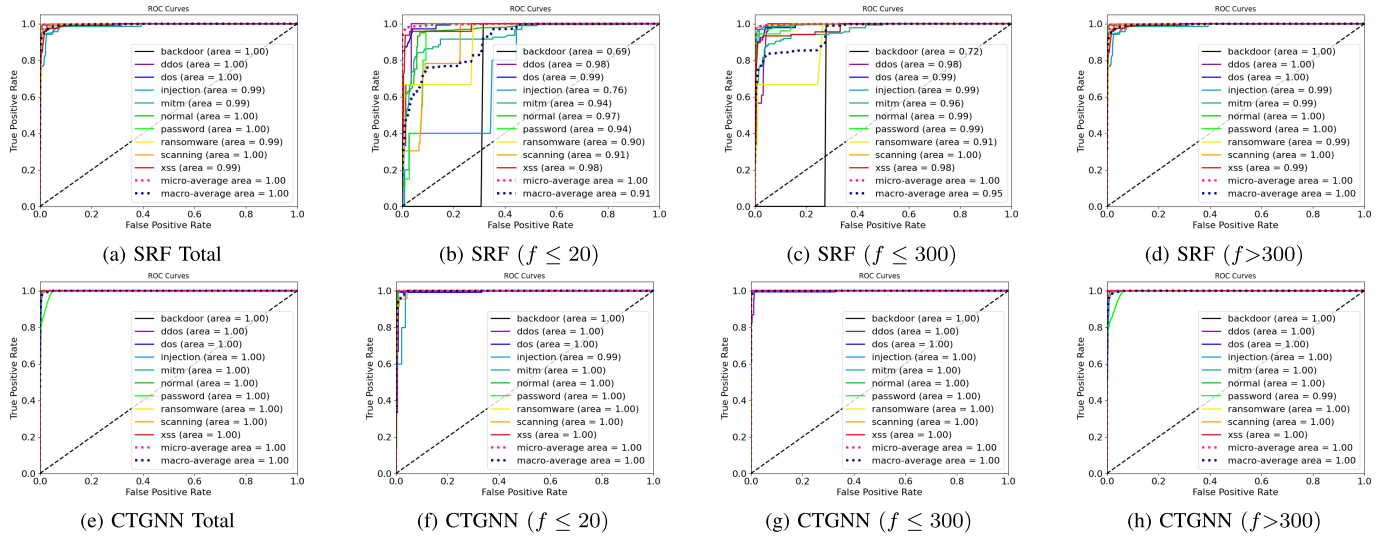


Fig. 6. For the Ton-IoT 2021 dataset, RoC curves are generated by SRF with communication frequency (b) $f \leq 20$, (c) $f \leq 300$, or (d) $f > 300$, as well as RoC curves on CTGNN with (f) $f \leq 20$, (g) $f \leq 300$, or (h) $f > 300$.

6) *Ablation Analysis:* We conduct additional ablation experiments to demonstrate the contributions of each component of our model to its overall performance. Essentially, we introduce historical representations (summarized by the interactive event update mechanism), spatiotemporal neighborhood information, and the time interval encoding process in the interaction behavior analysis. We design a simple and direct ablation experiment to verify the effect of each component on the overall performance. To eliminate the influence of the multihead attention mechanism on the proposed method, an MLP-based message aggregation approach is deployed to verify the role of each component in the interaction network analysis. To clarify how different components impact the interactive network analysis, we conduct ablation studies on the following message aggregation strategy variants: (1) a variant without neighborhood information, (2) a variant without the interactive event update mechanism, and (3) a variant without time interval encoding.

Fig. 7 presents the overall performance achieved in these confirmatory experiments compared with that of the MLP message aggregation-based CTGNNs and the base MLP. These CTGNN variants outperform the vanilla MLP model but perform slightly worse than the complete CTGNN. For instance, without neighborhood information aggregation, the

accuracy, precision, recall, and F1 score of the MLP message aggregation-based CTGNN drop by 1.26%, 7.64%, 4.22% and 5.98%, respectively. These results imply that (1) the introduced additional interactive event update mechanism, spatiotemporal neighborhoods, and time interval encoding components are all conducive to improving the performance achieved in the interaction behavior analysis. (2) The adopted computing components are not limited to a specific computing framework, such as multihead attention; in fact, these components are compatible with different feature aggregation schemes, including MLPs and transformers. (3) Neighborhood information has a greater impact on model performance than the other two computational components.

To further explore the expressive ability of our model, we conduct semisupervised learning with 20% of the labels sampled from the training sets of the ToN-IoT, CIC-Darknet and fraud payment datasets. Considering that the training set accounts for 80% of the total samples, the semisupervised learning process only uses 16% of the labels of the total samples. As shown in Table VI, when only 16% of the labels are utilized for these datasets, the results do not exhibit a large gap relative to those obtained when employing 80% of the labeled samples. The performance achieved with 16% of

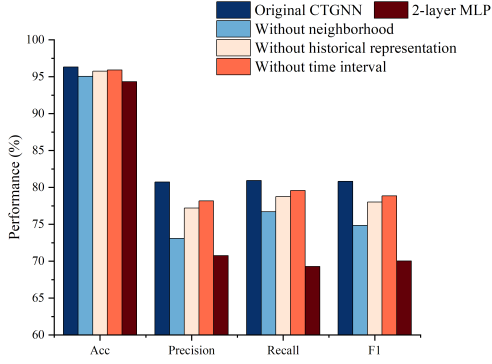


Fig. 7. Results of ablation experiments.

TABLE VI

ACCURACY AND F1 ACHIEVED WITH DIFFERENT LABEL PROPORTIONS FOR SEMISUPERVISED LEARNING ON 3 DATASETS

Experimental Setting	Dataset	ACC	F1
80% label sample	CICDarknet	99.72%	99.18%
16% label sample	CICDarknet	99.14%	97.47%
80% label sample	ToN-IoT	98.37%	98.36%
16% label sample	ToN-IoT	97.75%	97.73%
80% label sample	Fraud payment	95.99%	81.95%
16% label sample	Fraud payment	95.49%	78.61%

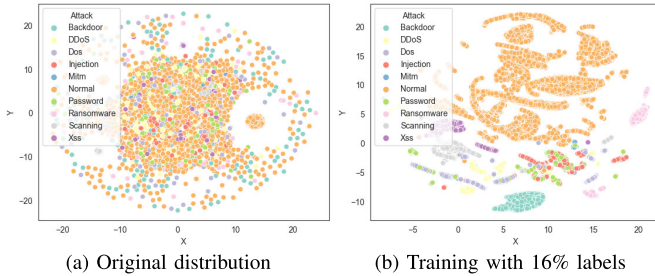


Fig. 8. Dimension-reduced visualizations of the embeddings.

the labeled samples decreases by only 0.58% and 0.45% on the CIC-Darknet and ToN-IoT datasets, respectively. In other words, when only 16% of the samples are labeled, our model already has sufficient expressive power for distinguishing between different types of behaviors.

Subsequently, we utilize the uniform manifold approximation and projection (UMAP) [27] algorithm to reduce the dimensionality of the high-level embedding representation obtained by our model. The visualization results in Fig. 8 a show the visualization results obtained with the original sample distribution of the ToN-IoT datasets. As shown in Fig. 8 b, the embedding representation of the network flows obtained with only 16% of the labeled samples already exhibits obvious classification boundaries. This visualization experiment presented in Fig. 8 further verifies that our model has sufficient expressive ability for distinguishing between different attack categories when few labeled samples are utilized.

7) *Summary*: Our contributions in this paper, as supported by the experiments above, can be summarized as follows.

(1) A novel network abnormality detection method that refines the concrete information interaction processes between each pair of entities is proposed, complementing substantial detailed behavior information. In the graph-based detection methods proposed in previous research [2], [4], [5], the extraction of temporal and spatial features is restricted to a fixed topology structure. In our work, via a customized CTGNN, we refactor the spatiotemporal correlation information as more granular entity interactions, which enhances the detection ability of the network and breaks through the fixed topology limitation. Experimental results show that this method works well on realistic scenarios, including intrusion detection, darknet analysis, and anti-fraud cases.

(2) We reveal the inherent shortcomings of traditional anomaly detection methods in the face of new node interactions and low-frequency or sporadic communication behaviors. Although traditional anomaly detection methods perform well on the overall test set, their good detection capabilities can only cover a small number of network entities that generate frequent interactions. The proposed CTGNN systematically utilizes the historical state, the local spatiotemporal information of neighbors and actual time decodings to maintain good performance in most of the tested node communication settings. To our knowledge, this is the first attempt to alleviate the negative impact of low-frequency communication in the field of anomaly detection.

(3) Ablation experiments demonstrate the effectiveness of each component in our proposed model. The key computing components include the CTG update mechanism, local spatiotemporal domain information, and time encoder, which exhibit good compatibility with various computational models and can strengthen their ability to detect anomalies. Furthermore, when only 16% of the total samples are labeled, our model achieves classification accuracies of 99.14%, 97.75%, and 95.49% on the CIC-Darknet2020, ToN-IoT, and Fraud Payment datasets, respectively, which are much higher than the accuracies achieved by the methods proposed in [4] and [26] when trained on 80% labeled samples. In an actual application scenario, these characteristics make our model competitive with other state-of-the-art methods via the imposition of a flexible computing overhead and weaker sample requirements.

VI. CONCLUSION

In various scenarios, the Continuous Temporal Graph Neural Network (CTGNN) consistently demonstrates its effectiveness in detecting anomalies across both binary and multiclass classification settings. Specifically, CTGNN achieves outstanding anomaly detection results, with F1 scores of 97.03%, 99.96%, 99.18%, and 81.95% on the UNSW-NB15, ToN-IoT, CICDarknet, and Fraud Payment datasets in binary classification settings. Moreover, our proposed method attains remarkable detection rates and F1 scores, surpassing 95%, across eight distinct traffic behavior types within the ToN-IoT dataset. Notably, CTGNN also achieves the highest macro-F1 score when applied to the UNSW-NB15 dataset.

Furthermore, manual verification results underscore the limitations of traditional detection methods, revealing oversights in the detection of behaviors generated by new access nodes

and behaviors characterized by low-frequency occurrences. CTGNN effectively addresses these challenges by harnessing the natural interaction processes of network entities in both temporal and spatial dimensions, thereby maintaining robust detection performance for these elusive behaviors.

Ablation experiments validate the versatility of the key computing components within CTGNN, showcasing their ability to enhance anomaly detection across various computational models. As we look ahead, our research will be dedicated to addressing the issue of strong data imbalance as well as further enhancing the interpretability of abnormal behavior detection methods. Furthermore, improving intrusion detection within the context of concept drift, characterized by shifts in traffic distribution and feature evolution, represents a challenge warranting considerable attention.

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